

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA5115 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5 , BL6)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

Code of Conduct:

I Nissanki Sri Venkata Lakshmi Kavya(192373036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

N.S.V.L.Kavya

Annexure A
PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

1. Write pseudocode to simulate a simplified TLS handshake process, including client request, server authentication using a digital certificate, session key generation, and secure data exchange.

PSEUDOCODE

ALGORITHM Simplified_TLS_Handshake

BEGIN

// Step 1: Client initiates connection

Client → Server: ClientHello

Client sends supported TLS versions,
supported cipher suites, and random value

// Step 2: Server responds

Server → Client: ServerHello

Server selects TLS version and cipher suite

Server generates and sends server random value

// Step 3: Server authentication

Server → Client: Digital Certificate

Client verifies:

- Certificate validity
- Certificate expiration
- Certificate issuer

- Server identity
- Digital signature

```
IF certificate verification fails THEN
    Display "Server authentication failed"
    Terminate connection
END IF
```

```
// Step 4: Session key generation
Client and Server perform secure key exchange
```

Client and Server independently generate
the same shared session key

```
// Step 5: Handshake completion
Client → Server: Finished
Server → Client: Finished
```

```
IF both Finished messages are verified THEN
    Display "TLS handshake successful"
ELSE
    Display "Handshake failed"
    Terminate connection
END IF
```

```
// Step 6: Secure data exchange
Client encrypts application data using session key
Client → Server: Encrypted Data
```

Server decrypts the data using session key

Server processes the data

Server encrypts response using session key

Server → Client: Encrypted Response

Client decrypts the response using session key

Display "Secure data exchange completed"

END

2. Develop pseudocode for a firewall rule engine that filters incoming packets based on IP address, port number, and protocol, and decides whether to allow or block traffic.

PSEUDOCODE

ALGORITHM Firewall_Rule_Engine

BEGIN

Define Firewall Rules

FOR each incoming packet DO

Extract:

Source_IP

Destination_IP

Port_Number

```
    Protocol
    Decision ← "BLOCK"

FOR each rule in Firewall_Rules DO

    IF Source_IP matches rule.Source_IP AND
       Port_Number matches rule.Port_Number AND
       Protocol matches rule.Protocol THEN

        IF rule.Action = "ALLOW" THEN
            Decision ← "ALLOW"
        ELSE IF rule.Action = "BLOCK" THEN
            Decision ← "BLOCK"
        END IF

        BREAK
    END IF

END FOR
IF Decision = "ALLOW" THEN
    Forward packet
    Log "Packet ALLOWED"
ELSE
    Drop packet
    Log "Packet BLOCKED"
END IF

END FOR
END
```

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient